

## 14.3 Multiple correlation

The population squared multiple correlation

$$\rho_{Y \cdot X}^2 = \frac{\boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}}{\sigma_Y^2}$$

is the largest squared correlation between  $Y$  and a linear combination of  $\mathbf{X}$  (Proposition 2.5.3). Section 9.5 studied its sample counterpart  $R^2$  for a fixed design (Theorem 9.5.1, Theorem 9.5.2). With random rows,  $R^2$  estimates  $\rho_{Y \cdot X}^2$ . This section finds its exact distribution under multivariate normality, and uses it to test  $\rho_{Y \cdot X}^2 = 0$ , to build an interval and to measure the bias.

Throughout,  $(\mathbf{X}_i^\top, Y_i)$ ,  $i = 1, \dots, n$ , are independent draws from a  $(k+1)$ -variate normal distribution with  $\boldsymbol{\Sigma}_{XX}$  positive definite and  $\sigma_Y^2 > 0$ , and  $n \geq k+2$ . By Eq. 3.5.2,  $Y_i = \alpha + \boldsymbol{\beta}^\top \mathbf{X}_i + e_i$  with independent  $e_i \sim N(0, \sigma^2)$  independent of the regressors, where

$$\sigma^2 = \sigma_Y^2(1 - \rho_{Y \cdot X}^2), \quad \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta} = \sigma_Y^2 \rho_{Y \cdot X}^2, \quad \text{so} \quad \frac{\boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}}{\sigma^2} = \frac{\rho_{Y \cdot X}^2}{1 - \rho_{Y \cdot X}^2}. \quad (14.3.1)$$

The ratio on the right is the population *signal-to-noise ratio*. We write  $\rho^2$  for  $\rho_{Y \cdot X}^2$  when no confusion can arise.

### 14.3.1 The sample multiple correlation as an estimator

Let  $\mathbf{S}_{XX}$ ,  $\mathbf{s}_{XY}$  and  $s_Y^2$  be the sample covariances, with any common divisor. The fit with an intercept has  $\text{SSR} = \hat{\boldsymbol{\beta}}^\top \mathbf{W} \hat{\boldsymbol{\beta}}$ , where  $\mathbf{W}$  is the centred cross-product matrix of the regressors and  $\hat{\boldsymbol{\beta}} = \mathbf{W}^{-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}})(Y_i - \bar{Y})$  (Theorem 6.6.2 with  $\mathbf{X}_2 = \mathbf{1}$ ). Dividing by SST gives

$$R^2 = \frac{\mathbf{s}_{XY}^\top \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}}{s_Y^2}, \quad (14.3.2)$$

the population formula with sample moments in place of population moments. Under normality it is more than a plug-in.

#### Proposition 14.3.1 — Maximum likelihood with normal rows

Suppose the sample covariance matrix of the rows is nonsingular. The maximum likelihood estimates of  $(\alpha, \boldsymbol{\beta}, \sigma^2)$  are the least squares estimates and  $\text{SSE}/n$ , and the maximum likelihood estimate of  $\rho_{Y \cdot X}^2$  is  $R^2$ .

*Proof.* The joint density of a row factorizes as  $f(\mathbf{x}; \boldsymbol{\mu}_X, \boldsymbol{\Sigma}_{XX}) f(y | \mathbf{x}; \alpha, \boldsymbol{\beta}, \sigma^2)$ , a normal density for the regressors times the conditional normal density of Theorem 3.5.2. The map from  $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  to  $(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_{XX}, \alpha, \boldsymbol{\beta}, \sigma^2)$  is one-to-one onto the set where  $\boldsymbol{\Sigma}_{XX}$  is positive definite and  $\sigma^2 > 0$ , with inverse  $\boldsymbol{\sigma}_{XY} = \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}$ ,  $\sigma_Y^2 = \sigma^2 + \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}$ ,  $\mu_Y = \alpha + \boldsymbol{\beta}^\top \boldsymbol{\mu}_X$ . So the log-likelihood is a sum of two terms with separate parameters, and each can be maximized on its own. The conditional term is the likelihood of a normal linear model

with model matrix  $[\mathbf{1}, \mathbf{X}_1]$ , maximized by least squares and  $\text{SSE}/n$  (Theorem 7.3.1).

In the marginal term the quadratic form is minimized over  $\boldsymbol{\mu}_X$  at  $\bar{\mathbf{x}}$ , leaving  $-\frac{n}{2} [\log \det \boldsymbol{\Sigma}_{XX} + \text{tr}(\boldsymbol{\Sigma}_{XX}^{-1} \mathbf{S})]$  with  $\mathbf{S} = \mathbf{W}/n$  positive definite. If  $\lambda_1, \dots, \lambda_k > 0$  are the eigenvalues of  $\mathbf{S}^{1/2} \boldsymbol{\Sigma}_{XX}^{-1} \mathbf{S}^{1/2}$ , then

$$\log \det \boldsymbol{\Sigma}_{XX} + \text{tr}(\boldsymbol{\Sigma}_{XX}^{-1} \mathbf{S}) - \log \det \mathbf{S} - k = \sum_{j=1}^k (\lambda_j - \log \lambda_j - 1) \geq 0,$$

with equality iff every  $\lambda_j = 1$ , that is, iff  $\boldsymbol{\Sigma}_{XX} = \mathbf{S}$ . By invariance of maximum likelihood under reparameterization, the estimate of  $\rho^2 = \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta} / (\sigma^2 + \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta})$  is  $\hat{\boldsymbol{\beta}}^\top \mathbf{S} \hat{\boldsymbol{\beta}} / (\text{SSE}/n + \hat{\boldsymbol{\beta}}^\top \mathbf{S} \hat{\boldsymbol{\beta}}) = \text{SSR}/(\text{SSE} + \text{SSR}) = R^2$ . ■

The rest of the section concerns the sampling distribution of this estimator.

### 14.3.2 The distribution of R-squared

#### Theorem 14.3.2 — Distribution of $R^2$ with normal rows

Under the assumptions above, let  $F = \frac{n-k-1}{k} \frac{R^2}{1-R^2}$  and  $\lambda = \rho^2/(1-\rho^2)$ .

- (a) Given the regressors,  $F \sim F(k, n-k-1, \gamma)$  with  $\gamma = \boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} / \sigma^2$ , where  $\mathbf{W}$  is the centred cross-product matrix of the regressors.
- (b) If  $\rho^2 = 0$ , then  $F \sim F(k, n-k-1)$  and  $R^2 \sim \text{Beta}(\frac{k}{2}, \frac{n-k-1}{2})$ , independently of the regressors. The overall  $F$  test of the regression is an exact size- $\alpha$  test of  $\rho^2 = 0$ , which under joint normality is the hypothesis that  $Y$  is independent of  $\mathbf{X}$ .
- (c) If  $\rho^2 > 0$ , then  $\gamma = \lambda V$  with  $V \sim \chi^2(n-1)$ , and for  $0 < c < 1$

$$\Pr(R^2 \leq c) = \mathbb{E} \left[ \Pr \left\{ F(k, n-k-1, \lambda V) \leq \frac{(n-k-1)c}{k(1-c)} \right\} \right]. \quad (14.3.3)$$

- (d) For each  $c \in (0, 1)$ ,  $\Pr(R^2 > c)$  is a strictly increasing function of  $\rho^2 \in [0, 1)$ .

*Proof.*

- (a) The centred regressor matrix has rank  $k$  with probability one, because  $\mathbf{W} \sim W_k(n-1, \boldsymbol{\Sigma}_{XX})$  (Lemma 14.2.3) is nonsingular when  $n-1 \geq k$  (the nonsingularity step in the proof of Lemma 14.2.4 needs only  $m \geq k$ ). Given the regressors, the model is a normal linear model with  $r = k+1$  and mean  $\alpha \mathbf{1} + \mathbf{X}_1 \boldsymbol{\beta}$ . By Theorem 9.5.2(a),  $F \sim F(k, n-k-1, \gamma)$  with  $\gamma = \|(\mathbf{M} - \mathbf{P}_1)(\alpha \mathbf{1} + \mathbf{X}_1 \boldsymbol{\beta})\|^2 / \sigma^2$ . Since  $(\mathbf{M} - \mathbf{P}_1)\mathbf{1} = \mathbf{0}$  and  $(\mathbf{M} - \mathbf{P}_1)\mathbf{X}_1 \boldsymbol{\beta} = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_1 \boldsymbol{\beta}$ , this is  $\boldsymbol{\beta}^\top \mathbf{X}_1^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{X}_1 \boldsymbol{\beta} / \sigma^2 = \boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} / \sigma^2$ .
- (b)  $\rho^2 = 0$  means  $\boldsymbol{\sigma}_{XY} = \mathbf{0}$ , so  $\boldsymbol{\beta} = \mathbf{0}$  and  $\gamma = 0$  for every design. By Lemma 14.1.1,  $F \sim F(k, n-k-1)$  independently of the regressors, and so is  $R^2 = kF/(kF + n - k - 1)$ , which is  $\text{Beta}(k/2, (n-k-1)/2)$  by Theorem 9.5.2(b). For jointly normal variables, zero covariance is independence (Theorem 3.4.1).
- (c) By the proof of Lemma 14.2.3,  $\mathbf{W} = (\mathbf{K} \mathbf{X}_1)^\top (\mathbf{K} \mathbf{X}_1)$ , where the  $n-1$  rows of  $\mathbf{K} \mathbf{X}_1$

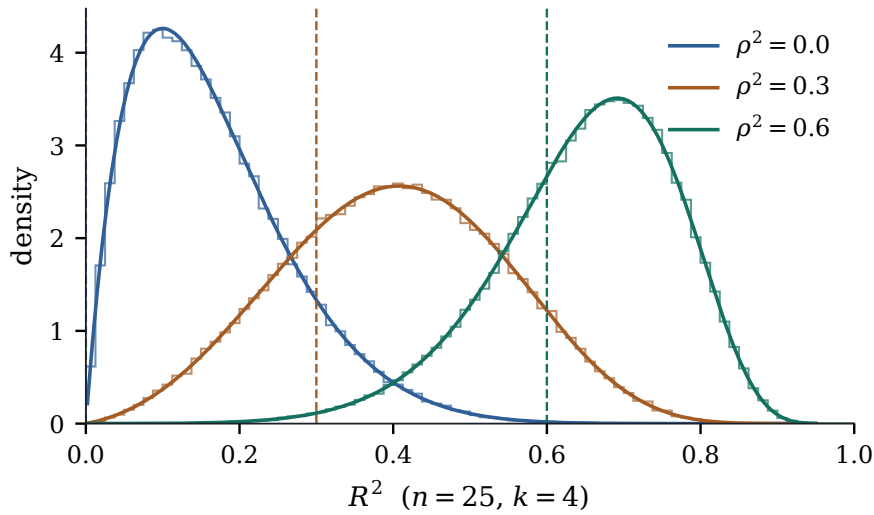


Figure 14.3.1: The law of  $R^2$  with normal rows,  $n = 25$ ,  $k = 4$ , for  $\rho^2 = 0, 0.3$  and  $0.6$  (dashed). Step outlines: simulation. Curves: Eq. 14.3.3.

are independent  $N_k(\mathbf{0}, \Sigma_{XX})$ . So  $\mathbf{KX}_1\boldsymbol{\beta}$  has independent  $N(0, \boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta})$  entries, and  $\boldsymbol{\beta}^\top \mathbf{W}\boldsymbol{\beta} = \|\mathbf{KX}_1\boldsymbol{\beta}\|^2 = (\boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta}) V$  with  $V \sim \chi^2(n-1)$ . By Eq. 14.3.1,  $\gamma = \lambda V$ . Since  $R^2 \leq c$  iff  $F \leq (n-k-1)c/(k(1-c))$ , taking expectations of the conditional probability from (a) gives Eq. 14.3.3.

- (d) Fix  $v > 0$  and  $f > 0$ . By Theorem 4.1.4,  $\Pr\{F(k, n-k-1, \lambda v) > f\}$  is strictly increasing in  $\lambda$ , and  $\lambda = \rho^2/(1-\rho^2)$  is strictly increasing in  $\rho^2$ . If  $\rho_1^2 < \rho_2^2$ , the conditional tail probabilities therefore differ by a positive amount for every value of  $V > 0$ , and so do their expectations.

■

Part (c) is Fisher's (1928) distribution of the multiple correlation coefficient, reached here without any integration. Given the design,  $R^2$  is a transformed noncentral  $F$  variable, and the design enters only through the single number  $\boldsymbol{\beta}^\top \mathbf{W}\boldsymbol{\beta}$ , which is a scaled chi-squared variable because the regressors are normal. Figure 14.3.1 shows the resulting densities for  $n = 25$  and  $k = 4$ , computed from Eq. 14.3.3 by averaging over a grid of quantiles of  $V$ .

#### Example. (How biased is R-squared?)

With  $n = 25$ ,  $k = 4$  and 100000 simulated samples, the averages of  $R^2$  are 0.1669, 0.4026 and 0.6500 for  $\rho^2 = 0, 0.3$  and  $0.6$ , against exact values 0.1667, 0.4026 and 0.6498. The exact values come from the formula  $E(R^2) = 1 - \frac{n-k-1}{n-1}(1-\rho^2) {}_2F_1(1, 1; \frac{n+1}{2}; \rho^2)$ , which follows from Fisher's (1928) distribution of  $R^2$  (see Olkin and Pratt 1958), where  ${}_2F_1$  is the hypergeometric function; it can also be computed from Eq. 14.3.3. The bias shrinks as  $\rho^2$  grows. The adjusted  $R^2$  of Eq. 9.5.1 averages 0.0003, 0.2831 and 0.5800: unbiased under the null (Proposition 9.5.3(e)), but biased *downwards* when  $\rho^2 > 0$ . Neither is unbiased everywhere. The standard deviations of  $R^2$  are 0.104, 0.146 and 0.118: with 25 cases, a single  $R^2$  says little about  $\rho^2$ .

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats
```

```
def r2_cdf(r, rho2, n, k, m=4000):
    """P(R^2 <= r) for iid normal rows: average the conditional noncentral F over V.
    """
    df2 = n - k - 1
    f = df2 * r / (k * (1 - r))
    if rho2 == 0:
        return stats.f.cdf(f, k, df2)
    V = stats.chi2.ppf((np.arange(m) + 0.5) / m, n - 1)      # equal-probability
    nodes
    return np.mean(stats.ncf.cdf(f, k, df2, rho2 / (1 - rho2) * V))
```

[Run this code online](#)

Part (d) says that the overall  $F$  test is unbiased and has power increasing in  $\rho^2$ , as a test of  $\rho^2 = 0$  against random-regressor alternatives. For  $n = 25$  and  $k = 4$  it rejects when  $R^2 > 0.364$ , and its power is 0.182, 0.377, 0.599 and 0.922 at  $\rho^2 = 0.1, 0.2, 0.3$  and 0.5. A population in which the regressors account for a fifth of the variance of the response is detected less than half the time.

### 14.3.3 An interval for the squared multiple correlation

Part (d) also makes an exact confidence interval possible. For observed  $R^2 = r$ , let  $\rho_L^2$  be the value at which  $\Pr(R^2 \leq r) = 0.975$  and  $\rho_U^2$  the value at which it equals 0.025 (with  $\rho_L^2 = 0$  if  $\Pr(R^2 \leq r) < 0.975$  already at  $\rho^2 = 0$ ). Because the probability decreases in  $\rho^2$ ,  $[\rho_L^2, \rho_U^2]$  covers the true  $\rho^2$  exactly when  $R^2$  lies between the 2.5% and 97.5% points of its law at that  $\rho^2$ , which has probability 0.95. This is the usual inversion of a family of tests (Proposition 12.1.2).

#### Example. (Violent crime in the states)

For the 50 states in 2009, regress the violent crime rate on the high-school graduation rate, the poverty rate, the percentage of single-parent households, the percentage of white residents and the urbanization rate. Then  $R^2 = 0.4607$ , the adjusted value is 0.3994, and  $F = 7.52$  on 5 and 44 degrees of freedom, with  $p$ -value  $3.6 \times 10^{-5}$ . Read with random regressors, the 95% interval for  $\rho^2$  is (0.175, 0.610). The interval is wide: five regressors and fifty cases leave much room for chance. It also treats the states as a random sample, which they are not; it describes hypothetical replications of a process generating rows like these, and here, as in Example (Poverty and graduation), no more can be claimed.

```
data = sm.datasets.statecrime.load_pandas().data.drop(index="District of Columbia")
regs = ["hs_grad", "poverty", "single", "white", "urban"]
y = data["violent"].to_numpy()
X = np.column_stack([np.ones(len(y))] + [data[c].to_numpy() for c in regs])
fit = sm.OLS(y, X).fit()
n_s, k_s = len(y), len(regs)
R2_obs = fit.rsquared
print(f"R^2 = {R2_obs:.4f}, adjusted {fit.rsquared_adj:.4f}, F = {fit.fvalue:.2f}, "
      f"p = {fit.f_pvalue:.2g}")
```

```
# 95% interval for rho^2: the values at which the observed R^2 is not extreme
def limit(prob):
    g = lambda rho2: r2_cdf(R2_obs, rho2, n_s, k_s) - prob
    return 0.0 if g(1e-12) < 0 else optimize.brentq(g, 1e-12, 0.999)

lower, upper = limit(0.975), limit(0.025)
print(f"95% interval for rho^2: ({lower:.3f}, {upper:.3f})")
```

[Run this code online](#)

*Remark* (Fixed or random?). Under a fixed design the expectation of  $R^2$  depends on the chosen regressor values (Exercise (B1)). With random rows  $R^2$  estimates  $\rho_{Y \cdot X}^2$ , which still depends on the regressor distribution: a narrower range gives a smaller  $\rho^2$ . The null law of the  $F$  test is the same in both readings.

## 14.3.4 Exercises

### 14.3.4.1 A. Check your understanding

#### Exercise A1.

A regression on  $k = 3$  random regressors from  $n = 40$  cases has  $R^2 = 0.25$ . Compute the overall  $F$  statistic and its degrees of freedom. Under the null hypothesis  $\rho^2 = 0$ , what are the mean and the distribution of  $R^2$ ?

### 14.3.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{R}$  be the sample correlation matrix of  $(Y, X_1, \dots, X_k)$ , assumed nonsingular, and let  $r^{YY}$  be the  $(1, 1)$  entry of  $\mathbf{R}^{-1}$ . Show that  $R^2 = 1 - 1/r^{YY}$ , and that  $1 - R^2 = \det \mathbf{R} / \det \mathbf{R}_{XX}$ , where  $\mathbf{R}_{XX}$  is the correlation matrix of the regressors.

*Solution.* Eq. 14.3.2 is unchanged when every variable is standardized, so  $R^2 = \mathbf{r}_{XY}^\top \mathbf{R}_{XX}^{-1} \mathbf{r}_{XY}$ . By Theorem 1.6.4, the leading entry of  $\mathbf{R}^{-1}$  is the inverse of the Schur complement  $1 - \mathbf{r}_{XY}^\top \mathbf{R}_{XX}^{-1} \mathbf{r}_{XY} = 1 - R^2$ , so  $r^{YY} = 1/(1 - R^2)$ . By Theorem 1.6.3(b),  $\det \mathbf{R} = \det \mathbf{R}_{XX} (1 - R^2)$ .

#### Exercise B2.

Show that  $R^2$  is unchanged if  $Y$  is replaced by  $aY + b$  with  $a \neq 0$ , or the regressors by  $\mathbf{B}^\top \mathbf{X} + \mathbf{c}$  with  $\mathbf{B}$  nonsingular. Show that  $\rho_{Y \cdot X}^2$  has the same invariance.

#### Exercise B3.

Show that  $R^2 \rightarrow \rho_{Y \cdot X}^2$  with probability one as  $n \rightarrow \infty$ , without assuming normality, pro-

vided the rows are independent and identically distributed with finite second moments and  $\Sigma_{XX}$  positive definite.

#### 14.3.4.3 C. Going deeper

##### Exercise C1.

With normal rows, show that the likelihood ratio statistic for  $\rho_{Y.X}^2 = 0$  against an unrestricted alternative is  $(1 - R^2)^{n/2}$ , so the likelihood ratio test is the overall  $F$  test. *Hint:* under the null the covariance matrix is block diagonal; use the proof of [Proposition 14.3.1](#) and [Exercise \(B1\)](#).

*Solution.* With  $\mathbf{S}$  the covariance matrix of the rows with divisor  $n$ , the maximized log-likelihood of a  $(k + 1)$ -variate normal sample is  $-\frac{n}{2}[\log \det \mathbf{S} + (k + 1)(1 + \log 2\pi)]$ , by the argument in the proof of [Proposition 14.3.1](#) applied to all  $k + 1$  variables. Under the null,  $Y$  and  $\mathbf{X}$  are independent, the likelihood factorizes, and the maximum is the same expression with  $\log(s_Y^2 \det \mathbf{S}_{XX})$  in place of  $\log \det \mathbf{S}$ . The likelihood ratio is therefore  $(\det \mathbf{S} / (s_Y^2 \det \mathbf{S}_{XX}))^{n/2} = (1 - R^2)^{n/2}$ , by Theorem 1.6.3(b) as in [Exercise \(B1\)](#). It is a decreasing function of  $R^2$ , hence of  $F$ .